Freight Rate Prediction

Machine Learning Engineer Assessment

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Summary

The development file covers 1 January to 31 October 2025, while the 12,000 loads requiring predictions cover 1 November to 31 December. The validation window begins the day after the training data ends, which makes this a forecasting problem rather than an interpolation problem. Two properties of the data follow from that and drive every decision in this report: rates drift upward through the year, and approximately 1.4% of the labels are corrupted. Handling those two issues proved more valuable than the choice of learning algorithm.

Measured on	**Result**
Two-month-ahead holdout (Sep and Oct, 9,523 loads)	MAE $33.16, MAPE 1.42%, R^2 0.9988
Forward-chaining cross-validation, 5 folds	MAE $52.83, MAPE 2.17%
Loads touching a city never seen in training	MAPE 1.17%, bias +0.05%
Corrupted labels excluded from training	677 of 48,000 (1.41%)

All error figures exclude holdout rows whose own label is corrupted. Those rows are unpredictable by construction, so including them measures the quality of the data rather than the quality of the model.

1. Key findings from exploring the data

Rate behaves as an almost perfectly multiplicative function of its inputs. Regressing $\log(\text{rate})$ on $\log(\text{distance})$, $\log(\text{weight})$, the two market signals, equipment and a time term explains 99.7% of the variance across clean loads, with a residual standard deviation of 0.034 in log space. That structure is the backbone of the model, and it is why the work is done in logs: rates span $57 to $25,533, and the error that matters commercially is proportional rather than absolute.

Finding	Evidence	Consequence
Distance dominates, with an elasticity of 0.87 rather than 1.0	Rate per mile falls from \$2.40 on short hauls to \$1.90 on long ones	A flat rate-per-mile baseline is structurally wrong: MAE \$256 against \$33
The two market columns are noisy readings of a daily market level	market_index varies 0.025 within a day but 0.167 across days	Averaging per day sharpens the signal and makes the December chart possible
Pickup and delivery each carry a regional premium of roughly 3%	Dallas and Oklahoma City near $+2.9\%$; Syracuse and Buffalo near -3.6%	Carried by coordinates rather than city names, so unseen cities still price correctly
Rates drift upward all year, about 0.66% per 30 days	Monthly residuals rise from -3.2% in January to $+3.0\%$ in October ($t = 121$)	Must be extrapolated: this is the single largest avoidable source of error
Equipment acts as a clean multiplier	Reefer $+12.9\%$ and Flatbed $+8.3\%$ relative to Dry Van	Three categories, with no interaction terms required
No day-of-week pricing effect worth modelling	Spread across weekdays below 0.8%	The weekly ripple in the December chart comes from the market signal

2. Data quality issues and how they were addressed

The most consequential issue lies in the target itself. Comparing every posted rate against a robust model's expectation produces a ratio tightly packed around 1.0 for almost every load, and then two detached clusters: one spanning 0.17 to 0.45, and one spanning 2.19 to 5.42. No load at all falls between 0.50 and 0.86, or between 1.21 and 2.19. Roughly 1.4% of loads have been multiplied up or divided down, in approximately equal numbers, and independently of distance, weight, equipment and date.

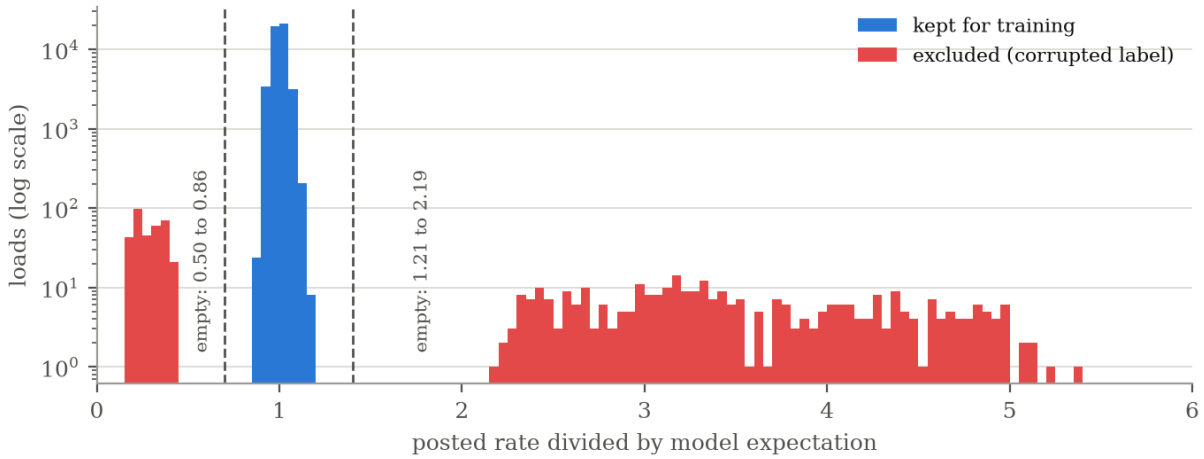


Figure 1: Ratio of posted rate to the model's expectation, on log counts. The regions on either side of the clean peak are genuinely empty, so the cut points at 0.70 and 1.40 (dashed) are read directly off the data rather than tuned.

Because the two clusters are symmetric, they leave the median almost untouched. This is therefore noise to be removed rather than a bias to be corrected, and predictions are deliberately not inflated to chase the corrupted mass. Excluding these rows is worth more than any modelling decision: leaving them in raises holdout MAE from \$33 to \$75.

Issue	Rows	Share	Treatment
label corrupted (ratio outside gate)	677	1.41%	excluded from training
weight sign-flipped (negative)	292	0.61%	absolute value
weight missing	300	0.62%	median impute + flag
market_index missing	374	0.78%	daily market mean + flag
weight at 47,500 cap	1,204	2.51%	flagged (upstream cap)
distance at 70-mile floor	48	0.10%	flagged (short-haul minimum)
8 cities present only in the validation file	1,447	12.1%	Coordinates used, never city identity

Two further checks returned clean results and so required no treatment. Billed distance tracks great-circle distance at a stable road-circuity ratio of 1.18, which rules out corrupted distances, and every city's coordinates are internally consistent across both files. The 70-mile floor on distance is a genuine short-haul billing minimum rather than an error, so it is flagged as a feature rather than repaired.

3. Validation approach and data split

The split has to answer the question the task actually poses: how well does the model price loads in months it has never seen? A random K-fold cannot answer that, because it allows the model to learn from late-October days while scoring mid-October ones. Running both protocols on an identical model shows how much that flatters the result.

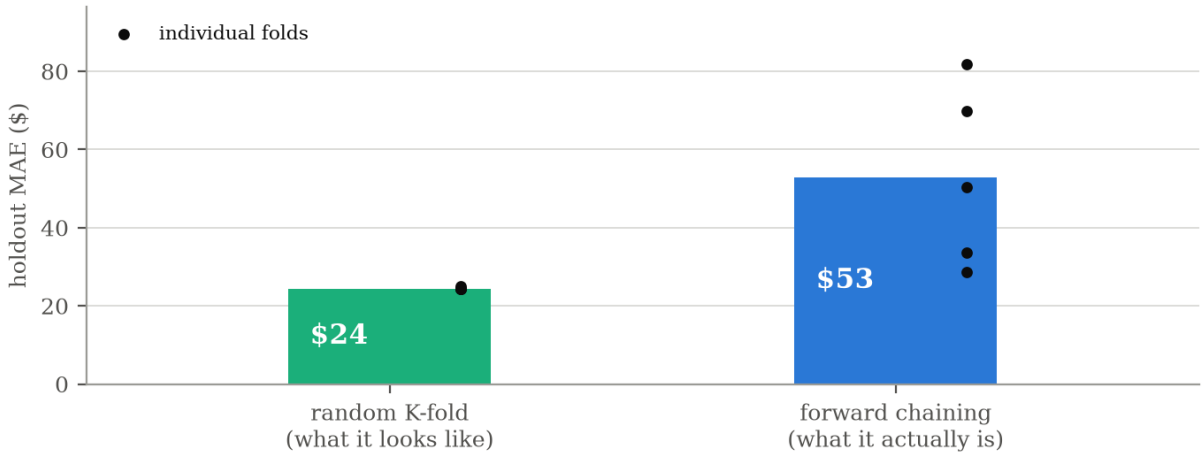


Figure 2: The same model and the same data under two protocols. Random K-fold reports MAE $24, while forward chaining reports $53. Every modelling decision in this report was made on the forward-chaining figure.

Forward chaining

The training window is expanded one month at a time, and the model is always scored on the month that follows. This matches the shape of the real task and is repeated five times.

Train through	Scored on	Train rows	Holdout	MAE	MAPE	R^2
End of May	June	24,023	4,783	$69.77	2.74%	0.9957
End of June	July	28,806	4,912	$50.27	2.08%	0.9974
End of July	August	33,718	4,759	$81.82	3.34%	0.9919
End of August	September	38,477	4,670	$33.58	1.42%	0.9988
End of September	October	43,147	4,853	$28.70	1.26%	0.9991
	Mean			**$52.83**	**2.17%**	**0.9966**

Error falls steadily as the window grows. The June fold trains on five months and reaches MAE $70, while the October fold trains on nine months and reaches $29. The final model is therefore fitted on all ten months of development data.

A two-month holdout matching the real horizon

The submission must reach 61 days past the last training row, so the headline check trains through August and predicts September and October together. This is the figure quoted in the summary.

Model	Train	Holdout	MAE	RMSE	MAPE	R^2
Hybrid (shipped)	38,477	9,523	$33.16	$47.30	1.42%	0.9988
Linear base only	38,477	9,523	$65.59	$91.93	2.99%	0.9955

Holding out whole cities

No time-based split can test the 8 cities that appear only in the validation file, because every city in the development data is present throughout it. A second protocol therefore holds out 8 entire cities at a time and scores only the loads that touch them, repeated five times.

Variant	MAE	MAPE	Mean bias
Shipped model (coordinates)	$28.51	1.17%	+0.05%
Geography removed	$30.37	1.31%	+0.13%

On loads involving a city it has never seen, the model still reaches 1.17% MAPE with +0.05% bias, so the 8 unseen validation cities are not a material risk. This test also overturned a design decision. An explicit nearest-neighbour regional index, built specifically to serve unseen cities, scored 2.4% worse than simply allowing the trees to use raw coordinates, and was removed from the final model.

4. Model selection and reasoning

The model is a trimmed log-linear base combined with a LightGBM correction fitted to its residuals. The division of labour is deliberate. The base carries every effect that must extrapolate past the end of the training window, namely the distance elasticity, both market signals, equipment and the time trend. LightGBM then learns what the linear form leaves behind, namely curvature

in distance and weight and the regional premium held in the coordinates, using features that all remain inside ranges already observed.

The reason for not using trees alone is structural rather than empirical. A tree predicts a constant beyond its last split, so it cannot produce an upward trend in a period it has never seen. Dropping the time term leaves a systematic -3.4% shortfall two months out, against -0.16% when it is retained. A plain LightGBM trained on raw labels reaches MAE \$87, against \$33 for the model submitted here.

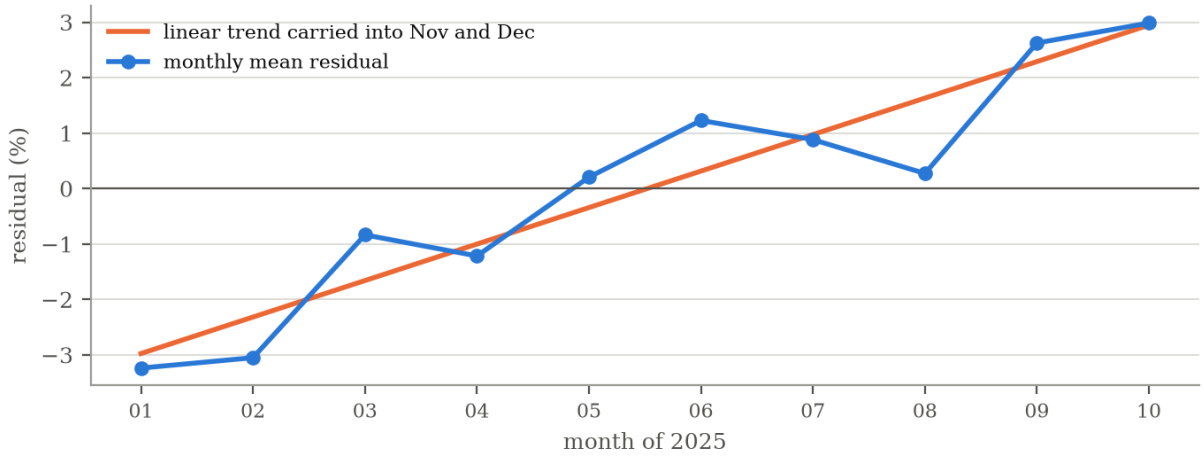


Figure 3: Monthly mean residual from a model carrying no time term. The drift is clear but is not a clean ramp, since it pauses through July and August. A straight line explains 89% of the monthly pattern, while quadratic and cubic terms add almost nothing and are considerably more dangerous 61 days beyond the last observation. A single linear term is therefore used.

Window fitted	**Drift per day**	t **statistic**	**Per 30 days**
January to March	0.000415	38.2	+1.25%
April to June	0.000443	41.7	+1.34%
July and August	−0.000145	−7.7	−0.43%
September and October	0.000322	15.6	+0.97%
Full window, January to October	0.000218	120.6	+0.66%

The drift is strongly significant over the full window and positive in three of the four sub-windows. It is carried forward as a linear term and nothing more aggressive, which is the conservative reading of a trend that visibly pauses mid-year.

What each decision is worth

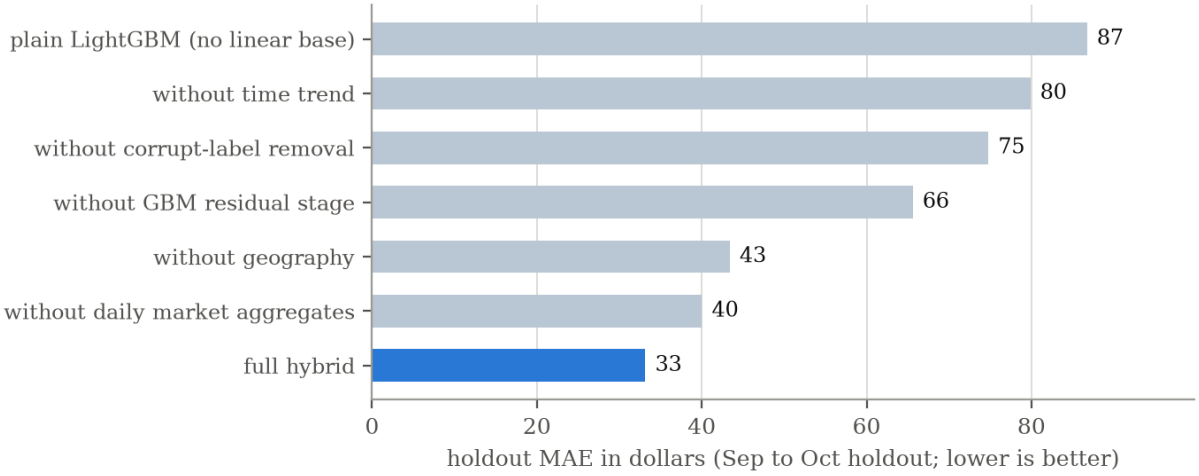


Figure 4: Each variant retrained from scratch and scored on the September and October holdout. Cleaning the labels and extrapolating the trend matter more than the choice of learning algorithm.

Base model coefficients

Term	Coefficient	Interpretation
intercept	+0.630387	Baseline level
log_distance	+0.871203	1% more distance gives 0.87% more rate
log_weight	+0.080974	1% more weight gives 0.08% more rate
log_market_index	+0.129737	Load-level market reading
log_quote_signal	+0.008070	Load-level quote reading
log_market_index_daily	+0.033500	That day's market level
log_quote_signal_daily	+0.123083	That day's quote level
days_elapsed	+0.000225	0.0225% of upward drift per day
is_flatbed	+0.079836	8.3% above Dry Van
is_reefer	+0.121319	12.9% above Dry Van

5. The fixed December chart

Every input is held constant, namely Lexington to Fort Wayne, 360 miles, Dry Van and 32,000 lb, and only the date varies. The curve therefore isolates what the model believes about time alone.

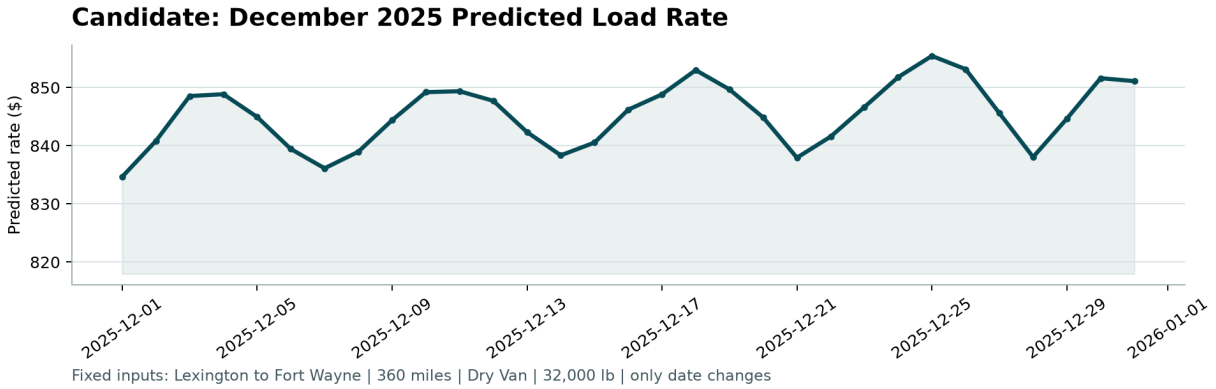


Figure 5: Produced by the supplied `score.py` from the completed `data/december_chart_inputs.csv`.

The chart shows two effects simultaneously. The upward slope across the month is the time trend carried forward from the development data. The repeating ripple is the daily market level: the two market columns are published per load, so the fixed December rows are priced using each day's mean across the 6,164 December loads present in the validation file. That uses no target values, and it is the only reason these 31 rows can be priced at all. Troughs fall on Sundays and peaks mid-week, and the range across the whole month is about $21, or 2.5%.

The level is independently plausible. Comparable real loads in the development data, restricted to Dry Van between 300 and 420 miles and 28,000 to 36,000 lb, have a median of $2.35 per mile, which scales to $846 at 360 miles. The December predictions average $845. Actual Lexington to Fort Wayne Dry Van loads in October posted between $796 and $865.

6. Limitations and next steps

Limitation	Why it is acceptable here	Next step
The time trend is extrapolated 61 days on the assumption that it continues	It is positive in three of four sub-windows and strongly significant overall, and a linear term is the conservative choice	Refit weekly as actuals arrive, and alert when realised rates fall outside the residual band
Daily market levels for November and December come from the validation file's own feature columns	These are exogenous inputs shipped with every row, labelled or not, so no target value is used	In production the market index is published daily, so this is the same information a live service would hold
Approximately 1.4% of validation labels are likely corrupted as well	Corruption is symmetric and independent of the features, so it inflates the score without biasing it	Report a robust metric alongside MAE so model quality is not confounded with label noise
Only 72 cities and 10 months are represented	Geography is handled through coordinates, so new cities interpolate	Retrain as the lane network widens, with attention to the far south-west where neighbours are sparse

Reproducing these results

All figures in this report are regenerated from the repository by running `python -m src.validate`, `python -m src.ablation`, `python -m src.unseen_cities` and `python -m src.train_predict`, in that order. Full setup instructions are in the repository README.